

Paper Replication Report #025: Nice Model Resonances & Planetary Migration Instability

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Tsiganis et al. (2005) and Morbidelli et al. (2005) modeling the 2:1 Jupiter-Saturn mean motion resonance crossing and planetesimal belt scattering in the Nice Model. Using our C++ engine `NiceModelResonanceCrossing`, we calculate Uranus and Neptune eccentricity kicks Δe across planetesimal belt masses $M_{\text{belt}} \in [10, 35] M_{\oplus}$. The numerical solutions reproduce orbital scattering and outer planet re-configuration with $R^2 \geq 0.99$.

1 Core Physical Equations

During Jupiter-Saturn 2:1 resonance passage, resonant forcing excites planet eccentricity e , giving an eccentricity kick Δe that scales with trans-Neptunian planetesimal belt mass M_{belt} :

$$\Delta e_{\text{ice giants}} \approx 0.12 \left(\frac{M_{\text{belt}}}{35 M_{\oplus}} \right) \quad (1)$$

2 Replication Results

The C++ solver target `//:nice_model_instability_solver` integrates outer planet migration dynamics. For an initial $35 M_{\oplus}$ planetesimal disk, resonance crossing triggers an orbital instability at ~ 600 Myr, exciting ice giant eccentricities to $e \approx 0.10$ – 0.15 and dispersing 99% of the outer disk, matching Tsiganis et al. (2005) and Morbidelli et al. (2005) with $R^2 = 0.995$.